ASP.NET MVC Dependency Injection - ████ ████ ██████████

██████ 1: ██████

███ Dependency Injection ██ DI ██ ███ ████████ ██ ASP.NET Core. ████ ██ ███████ ███████ AddScoped ██ Constructor Injection ████ ███ ████████ ████ ████ DI ██████. ██████ ██ ███ ██████ ██ ███ ███████ ██ ██████ ███ ███████ ██████ ██ ███████ ██ Interview.

██████ 2: ██ ██ Dependency█

███ ███ Controller ██████ Repository ███ ████ ████████ ███ Repository ██████ Dependency ██Controller. ████ ████ Repository ██████ DbContext ███ DbContext ██████ Dependency ██Repository. ██ Object ███████ Object ███ ███ ████ ████ Dependency.

██████ 3: ███ ████████ ████ DI

DepartmentController ████ ███████ DepartmentRepository ████████ new. ██ DepartmentRepository ████ ███████ MVCAppDbContext ████████ new. ████ ██████ ██ Class ██████ ██ ██████ ███████ ████ ██████ ████.

██████ 4: ██████ Tight Coupling

██████ ████ Controller ██████ new DepartmentRepository() ████ ████ ████ ███ ███████ ████████. ███ ████ ██████ ██ ███ Constructor ██ ██████ ████████ ███ ██████ ███ ██████ ██████ ██████. ████ ███ ████████ ███ ████████ ██████ ████████ ████.

██████ 5: ██████ ██ ██████ Dependency ██████ ███████

████ ██████████ ██████. Controller ██████ Repository ██████. ████████ ████ ██ ████ ████████████ ██ ██ ██ ███████ ████ ████████████ ██████. DI ██ ████ ██████████ ██████ ████ ████████ ████████.

██████ 6: ████████ ██████████ ███ DI

██████ ██ ██ ████ Controller: new DepartmentRepository() ████ ███: ███ ██████ IDepartmentRepository. ██ ████ ██████ ██████.

██████ 7: ████ ████ Injection

Injection ████ ███ ██ ██████ ████████████ ███████ ██ ███████. ██ ██ Repository ███ ████████ ██Controller ██████ ██ ██ ███████ ██████.

██████ 8: Constructor Injection

[illegible] [illegible] [illegible] DI. [illegible] [illegible] [illegible] [illegible] Constructor. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] Constructor.

[illegible] 9: Interface [illegible] [illegible]

[illegible] [illegible] [illegible] DepartmentRepository [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] IDepartmentRepository [illegible] [illegible] [illegible] [illegible]. [illegible] Class [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] Controller.

[illegible] 10: Depend on Abstraction

[illegible] [illegible] SOLID [illegible]: Depend on Abstractions not Concrete Implementations. [illegible] [illegible] [illegible] Interface [illegible] [illegible] Class [illegible].

[illegible] 11: DI Container

ASP.NET Core [illegible] [illegible] DI Container [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] Objects [illegible] [illegible] [illegible] [illegible].

[illegible] 12: [illegible] [illegible] Program.cs

AddScoped() [illegible]: [illegible] [illegible] [illegible] [illegible] IDepartmentRepository [illegible] [illegible] DepartmentRepository [illegible].

[illegible] 13: [illegible] [illegible] Request

1- [illegible] [illegible] Request. 2- MVC [illegible] [illegible] Controller. 3- [illegible] [illegible] Controller [illegible] IDepartmentRepository. 4- [illegible] [illegible] Container. 5- Container [illegible] Repository. 6- [illegible] [illegible] Repository DbContext [illegible] [illegible]. 7- [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible] 14: DbContext Injection

[illegible] [illegible]: new MVCAppDbContext() [illegible] [illegible]: [illegible] MVCAppDbContext [illegible] Constructor. [illegible] [illegible] ASP.NET [illegible] [illegible] [illegible] [illegible].

[illegible] 15: Bug [illegible]

[illegible] [illegible] [illegible] Field [illegible] _dbContext [illegible]Parameter [illegible] dbContext [illegible]: _dbContext = dbContext; [illegible] dbContext = dbContext [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible] 16: Scoped

Scoped [illegible] Object [illegible] [illegible] Request. [illegible] [illegible] Controller [illegible]Service [illegible] Repository [illegible] [illegible] Request [illegible] [illegible] [illegible] [illegible].

[illegible] 17: Transient

Transient [illegible] [illegible] Object [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

■■■■■ 18: Singleton

Singleton ■■■■ ■■■■ ■■■■ ■■■■■ ■■■■ ■■■ ■■■■■■■■. ■■ ■■■■■■■■■ ■■■ Requests ■■■■■■■■■■ ■■■ ■■■■■■■.

■■■■■ 19: ■■■■■ DbContext ■■■■■■■ Scoped

Entity Framework ■■■■■■ ■■■ ■■■■ ■■■■■■■■■. ■■■■ DbContext ■■■■■ ■■■ Request ■■■■ ■■■■■■ ■■■■■■ ■■ ■■■■■■■■ ■■■■ Tracking.

■■■■■ 20: Unit Testing

■■■■ DI ■■■■ ■■■■■■■■ Repository. ■■ DI ■■■■ ■■■■■■ Mock ■■■■■■■■ ■■Controller ■■■■■■■. ■■■■ ■■■■■■ ■■■■■■ DI ■■■■ ■■■■ ■■ ■■■■■■■■■■■■.

■■■■■ 21: ■■■■■■■ ■■■■■■

■■■■ DI: Controller -> new Repository -> new DbContext ■■ DI: Controller <- Repository Repository <- DbContext Container ■■ ■■■■■■■■ ■■ ■■■■■■■■.

■■■■■ 22: ■■■■■■ ■■■■■■■■■

■■ ■■■■■■ ■■■■ Scoped ■ Singleton■ ■■■■■■ ■■■■■■■ Interface■ ■■ ■■■■■■ DI Container■ ■■ ■■■■■■ ■■■■ Dependency ■ Dependency Injection■ ■■■■■■ ■■ ■■■■■■■ new ■■■■ Repository■

■■■■■ 23: ■■■■■ ■■■■■■

Dependency = ■■■■ ■■■■■■■ ■■■■■■■. Injection = ■■■■■■ ■■■■ ■■■■■■ ■■ ■■■■■■■. DI Container = ■■■■■■ ■■ ■■■■■■■■. Interface = ■■■■. Repository ■ DbContext ■■■■■■■ Scoped. DI ■■ ■■■■■ ■■■■ Requests ■■■■ ■■■■■■■. DI ■■■■■ ■■■■■■■■ ■■■■■■■■■■ ■■■■■■ Coupling.